

## Unit – I: Introduction to Vibration

Types of vibrations, Definitions, Simple Harmonic Motion (S.H.M.), Work done by harmonic force, Principle of super position applied to SHM, Beats, Fourier theorem and problems.

Vibration is defined as the to and fro motion of the disturbed structure from its equilibrium position. There are many ways in which vibration can be seen or experienced. The simple way to understand vibration can be made by taking musical instruments as example.

- Drums, flute, whistle, saxophone etc. are all sources of vibration. It can be again stated, these vibrations can be sometimes pleasant to our ears and sometimes irritating. Hence vibration became point of interest for scientists.

In other words, it can be stated that, all bodies having mass and elasticity are capable of vibration. The mass is inherent of the body and elasticity causes relative motion among its parts.

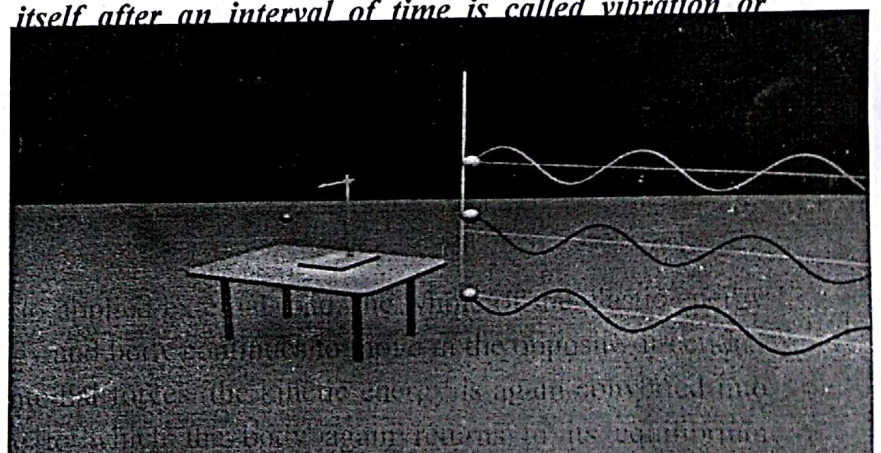
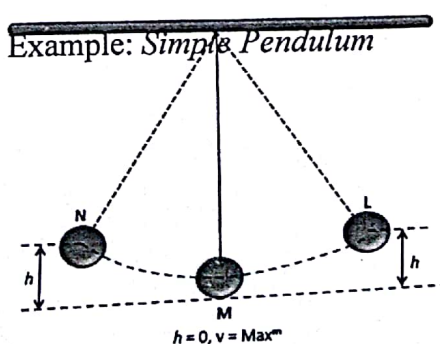
### BASIC UNDERSTANDING IN VIBRATION

- It is well known fact that, all matter is made up of particles. Which means to say that it has a definite shape, size and mass.
- If this closely packed particles are disturbed by some external loading, the mass of one body tries to impart its energy to the neighbouring particles. Due to which the particles are displaced by the application of external forces.
- At the same time, the internal forces in the form of elastic energy tries to bring back the body to its original position.

#### **Energy transformation**

- At equilibrium position, due to applied external load, the whole of the elastic energy is converted into kinetic energy and body continues to move in the opposite direction.
- Simultaneously, due to the internal forces, the kinetic energy is again converted into elastic or strain energy, due to which the body again returns to its equilibrium position.
- Due to this, vibration is continuous and exchange of energy takes place repeatedly.

*Thus, any motion which repeats itself after an interval of time is called vibration or oscillation.*



## Simple Definitions

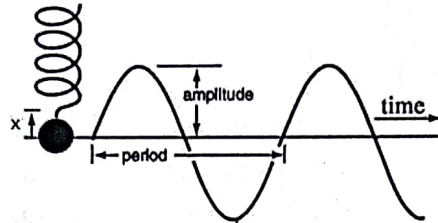
**Periodic Motion:** A motion which repeats itself after equal interval of time.

**Time Period:** Time taken to complete one cycle.

**Frequency:** Number of cycles per unit time.

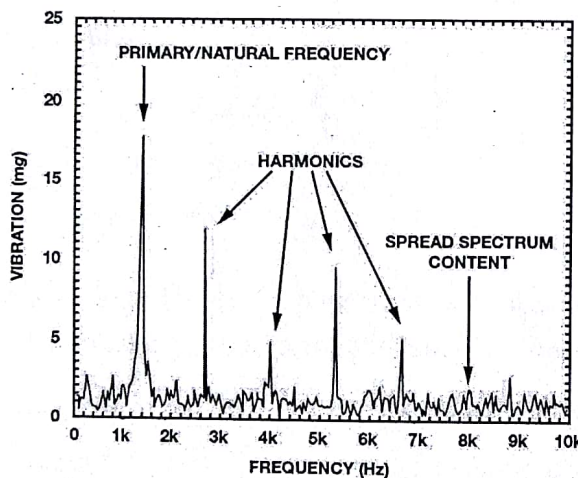
**Amplitude:** The maximum displacement of a vibrating body from its equilibrium position.

Periodic Motion Graph

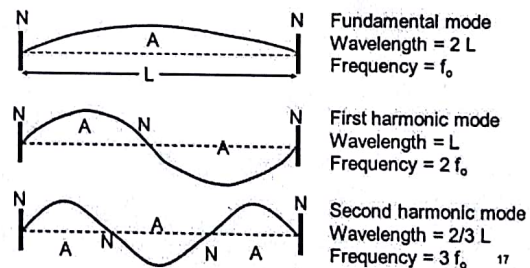


**Natural frequency:** If the structure is disturbed by giving initial displacement alone and with no external force, then the vibration is called as free vibrations. And the frequency at which it is vibrating is called as natural frequency expressed in Hertz or rad/sec.

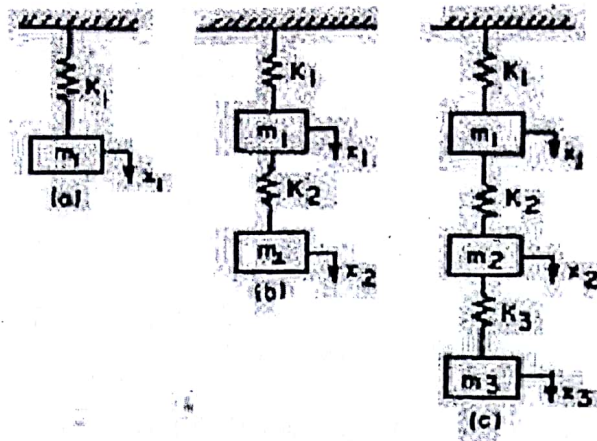
**Fundamental Mode of Vibration:** In a vibrating system, the mode having lowest natural frequency is called the fundamental mode of vibration. Modes of vibration



- Nodes N  $\rightarrow$  the string does not move
- Antinodes A  $\rightarrow$  string has maximum amplitude



**Degree of freedom:** The minimum number of independent coordinates required to specify the motion of a system at any instant is known as degrees of freedom of the system.



In single degree of freedom system, there is only one independent coordinates used to specify the configuration or motion of the system. Similarly, for two and three degree of freedom system, there are two and three coordinates required to represent the motion respectively.

*Infinite degrees of freedom example: A Cantilever beam.*

**Simple Harmonic Motion (SHM):** The periodic motion of a body, to and fro about its equilibrium position is called Simple Harmonic Motion.

The motion is periodic, and in reality its acceleration is always directed towards the mean position and is proportional to its distance from mean position. SHM is represented by the equation,

$$x = A \sin(\omega t)$$

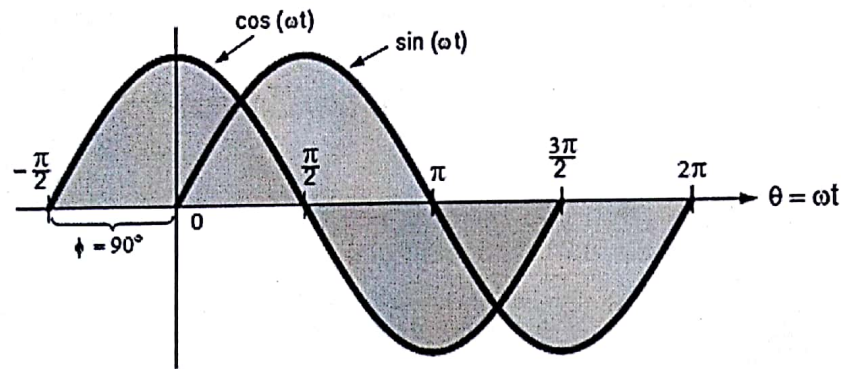
$$\dot{x} = A\omega \cos(\omega t)$$

$$\ddot{x} = -A\omega^2 \sin(\omega t) = -\omega^2 x$$

**Damping:** The resistance offered against the motion of vibrating body is known as damping. The vibration associated with this resistance are known as damped vibrations.

**Phase Difference:** It is the difference, expressed in degrees or radians, between two waves having the same frequency and referenced to the same point in time.

Two oscillators that have the same frequency and no phase difference are said to be in phase. Two oscillators that have the same frequency and different phases have a phase difference, and the oscillators are said to be out of phase with each other.



**Resonance:** When the external excitation frequency is equal to the natural frequency of vibrating body, the amplitude of vibration becomes excessively large. This phenomenon is known as resonance.

**Mechanical System:** The system consisting of mass, stiffness and damper is known as mechanical system.

**Discussion: Difference between Continuous and Discrete System**

### Vibrating system

As discussed, in a vibrating system there is always exchange of energy from one form to another. For a mechanical system having spring, mass and damper as a basic component,

1. Energy is stored by **mass** in the form of kinetic energy.  $\left(\frac{1}{2}m\dot{x}^2\right)$
2. In the **spring**, energy is stored in the form of potential energy.  $\left(\frac{1}{2}kx^2\right)$
3. Also energy is dissipated in the **damper** in the form of heat energy, which opposes the motion of the system.

If the system is disturbed by external excitation, the energy enters the system. This excitation energy disturbs the mass from its mean position and the mass goes up and down from its mean position. During which kinetic energy is converted to potential energy and potential energy into kinetic energy. This process continues and makes the system to vibrate. Simultaneously damping force acts on the mass and opposes its motion. Due to which certain amount of energy is dissipated as heat and the free vibrations die out and attains static equilibrium position.

The equation of motion for a vibrating system having free vibration can be written as

$$M\ddot{x} + C\dot{x} + Kx = 0$$

### Methods of Vibration analysis

The equation of motion for a vibrating system can be obtained by different methods

1. **Energy Method:** For the system having vibration, the sum of the energies associated with the system is constant.

Kinetic Energy + Potential Energy = Constant

(KE + PE) = constant

$$\frac{d}{dt} \left( \frac{1}{2} m \dot{x}^2 + \frac{1}{2} k x^2 \right) = 0$$

$$m \ddot{x} + k x = 0$$

$$m \ddot{x} + k x = 0$$

This is the equation of motion considering no damping.

If the motion produced by the vibrating system is assumed to be Simple Harmonic, then it is given as

$$x = A \sin(\omega t)$$

$$\text{So, } \ddot{x} = -A \omega^2 \sin(\omega t)$$

Substituting the above in the obtained equation of motion, we get

$$-m A \omega^2 \sin(\omega t) + k A \sin(\omega t) = 0$$

Thus

$$\omega = \sqrt{\frac{k}{m}} \text{ rad/sec}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \text{ Hz}$$

2. **Rayleigh's Method:** This method is the extension of Energy method and it states that the total energy of the vibrating system is equal to the maximum potential energy.

$$\text{KE} = \text{PE}$$

$$\frac{1}{2} m \dot{x}^2 = \frac{1}{2} k x^2$$

$$m(A\omega)^2 = kA^2$$

$$m\omega^2 = k$$

$$\omega = \sqrt{\frac{k}{m}} \text{ rad/sec}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \text{ H}$$

3. **Equilibrium Method:** According to this method, the algebraic sum of the forces and moments acting on the system must be zero. If the external forces acting on the system is  $F$ , spring force  $kx$ , damping force  $c\dot{x}$  and inertia force  $m\ddot{x}$ , then the equation of motion is written as

$$m\ddot{x} + c\dot{x} + kx = F$$

## **TYPES OF VIBRATION**

### **1. Free and forced vibration (*Based on the input disturbance*)**

If the system is disturbed by some initial disturbance, immediately after the input, the vibration continues on its own. This type of vibration is known as free vibration.

Eg: Simple Pendulum

If the Structure is disturbed by the influence of some external excitation, then such type of vibration is called as forced vibration.

Eg: Electric bells

### **2. Linear and Nonlinear Vibrations (*Based on the nature of response*)**

In a vibrating system, if the spring, mass and damper behave in a linear manner, then the vibrations caused also will be linear in nature. Linear vibrations are governed by linear differential equations and they follow the law of superposition.

On the other hand if the vibrating components behave non-linearly, the vibration is called as non-linear. For very large amplitude of vibrations, linear vibrations become non-linear.

### **3. Damped and Undamped Vibrations (*Based on the internal resistance to motion*)**

If the motion of the vibrating system is opposed by the damper, then the energy will be dissipated in friction generating heat. This type of vibration is called damped vibration.

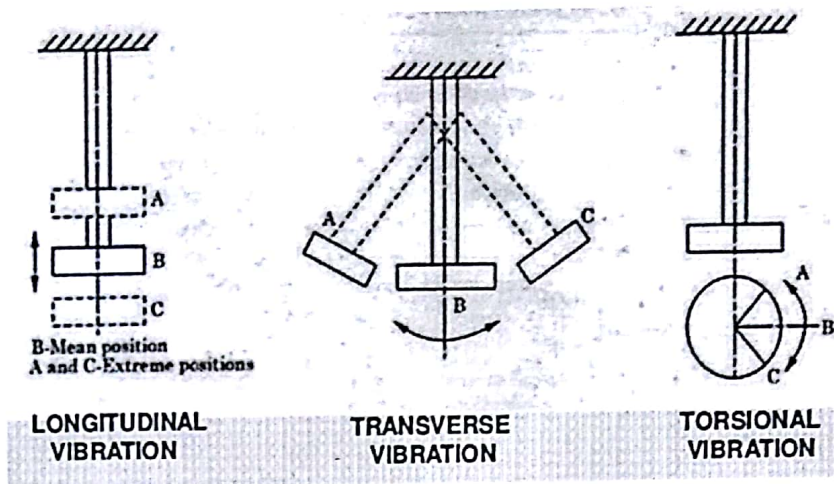
On the other hand, the system having no damper is known as undamped system. In reality no such system exists.

### **4. Deterministic and random Vibration (*Based on the amount of external excitation*)**

If in the vibratory system the amount of external excitation is known in magnitude, it causes deterministic vibration. Contrary to it the non-deterministic vibrations are known as random vibrations.

### **5. Longitudinal, Transverse and Torsional Vibrations**

It can be seen from the figure, that if the motion of the vibrating structure has mass moving, up and down parallel to the spindle axis, it is said to be having Longitudinal vibrations.

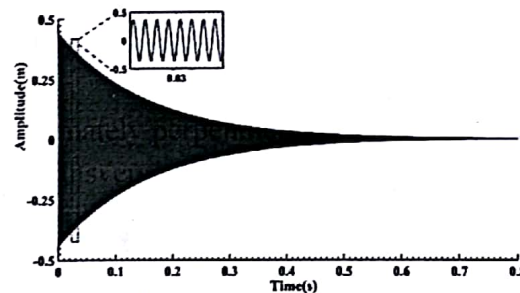


If the shaft moves approximately perpendicular to the axis of shaft, then the vibration so caused are known as Transverse.

On the other hand, if the spindle gets alternatively twisted and untwisted on account of vibratory motion of the suspended disc, it is called to undergoing torsional vibrations.

## 6. Transient Vibration

For an ideal system, the free vibrations does not die out but will have indefinitely continuous vibrations assuming, having no damping. But in a real system, the vibrations die out due to damping and vanishes ultimately. Such type of vibrations are called as Transient vibrations.



## Simple Harmonic Motion

The motion which repeats itself after equal interval of time is known as periodic motion. The equal time intervals are called as time period.

Let us consider a motion which repeats after every  $\frac{2\pi}{\omega}$  time and having the type

$x_1 = A_1 \sin \omega t$ , where  $\omega$  = Natural frequency,  $x_1$  = displacement and  $A_1$  = amplitude

Generally harmonic motions are represented by circular sine and cosine function. All harmonic motions are periodic but vice- versa is not always true.

*Note: In a simple harmonic motion, acceleration is always proportional to its displacement and directed towards a particular fixed point.*

**Superposition:** When Harmonic motions of same period are added, the resultant harmonic motion also will have same time period.

### Addition of Harmonic motion

Suppose if we add, simple harmonic motions having same frequency, the resultant motion we get will also be harmonic.

To Prove: *Addition of two harmonic motion yields harmonic resultant.*

$$x_1 = A_1 \sin \omega t$$

$$x_2 = A_2 \sin(\omega t + \phi)$$

Adding the above two harmonic motions, we get the resultant motion as,

$$x = x_1 + x_2 = A_1 \sin \omega t + A_2 \sin(\omega t + \phi)$$

But we all know that,  $\sin(A+B) = \sin A \cos B + \cos A \sin B$ , therefore on expanding, we get

$$x = \sin \omega t (A_1 + A_2 \cos \phi) + A_2 \cos \omega t \sin \phi$$

Let us assume,  $(A_1 + A_2 \cos \phi) = A \cos \theta$  and  $A_2 \sin \phi = A \sin \theta$ , therefore the above equation is written as,

$$x = A \sin \omega t \cos \theta + A \cos \omega t \sin \theta$$

$$x = A \sin(\omega t + \theta)$$

The above equation proves that the resultant displacement is also simple harmonic motion of amplitude A and phase ' $\theta$ '.

**To find the resultant Amplitude:** Squaring and adding the assumed values, we have

$$A^2 (\cos^2 \theta + \sin^2 \theta) = (A_1 + A_2 \cos \phi)^2 + A_2^2 \sin^2 \phi$$

$$A^2 = (A_1 + A_2 \cos \phi)^2 + A_2^2 \sin^2 \phi$$

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \phi \text{ and}$$

$$A = \sqrt{(A_1^2 + A_2^2 + 2A_1A_2 \cos \phi)}$$

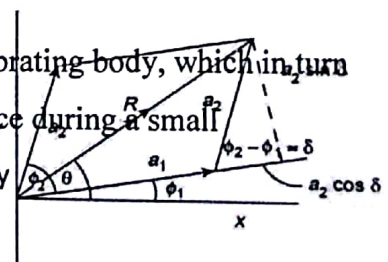
The phase difference is determined from the same assumed equation,

$$\text{i.e. } \tan \theta = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}, \text{ therefore}$$

$$\theta = \tan^{-1} \left( \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi} \right)$$

### WORK DONE BY HARMONIC FORCE

Let us consider a harmonic force  $F = F_0 \sin \omega t$  to be acting on a vibrating body, which in turn has a motion  $x = x_0 \sin(\omega t - \phi)$ . The work done by the applied force during a small displacement  $dx$  is  $F \cdot dx$ . Hence the work done in one cycle is given by



$$W = \int_0^T F \frac{dx}{dt} dt = \int_0^T \left[ F_0 \sin \omega t \frac{d}{dt} x_0 \sin(\omega t - \phi) \right] dt$$

$$W = \int_0^T F_0 \sin \omega t x_0 \omega \cos(\omega t - \phi) dt$$

$$W = x_0 F_0 \omega \int_0^T \sin \omega t \cos(\omega t - \phi) dt$$

$$W = x_0 F_0 \omega \int_0^T \left[ \frac{\sin 2\omega t \cos \phi}{2} + \frac{\sin \phi (1 - \cos 2\omega t)}{2} \right] dt$$

Substituting  $T = \frac{2\pi}{\omega}$ , we get

$$W = \pi F_0 x_0 \sin \phi$$

From the above equation, it is clear that if  $\phi = 0$ , then the work done will be zero. Which means force and displacement should not be in phase to get the work done.

## Unit - I - Problems

- ① Add the following harmonic motions analytically and check the solution graphically.

$$x_1 = 4 \cos(\omega t + 10^\circ)$$

$$x_2 = 6 \sin(\omega t + 60^\circ)$$

Solution: Frequency is same for both  $x_1$  &  $x_2$   $\therefore$   $x = A \sin(\omega t + \alpha)$

$$x = x_1 + x_2$$

$$A(\sin \omega t \cos \alpha + \cos \omega t \sin \alpha) = 4 \cos(\omega t + 10^\circ) + 6 \sin(\omega t + 60^\circ)$$

$$\Rightarrow 4 \cos \omega t \cos 10^\circ - 4 \sin \omega t \sin 10^\circ + 6 \sin \omega t \cos 60^\circ + 6 \cos \omega t \sin 60^\circ$$

$$\therefore \sin \omega t (A \cos \alpha) + \cos \omega t (A \sin \alpha) = \sin \omega t (-4 \sin 10^\circ + 6 \cos 60^\circ) + \cos \omega t (4 \cos 10^\circ + 6 \sin 60^\circ)$$

$$= \sin \omega t (-0.6943 + 3) + \cos \omega t (3.9392 + 5.1961)$$
$$= \sin \omega t (2.305) + \cos \omega t (9.135)$$

By equating corresponding Co-efficients of  $\cos \omega t$  and  $\sin \omega t$  on both sides

$$A \cos \alpha = 2.305 \text{ and } A \sin \alpha = 9.135$$

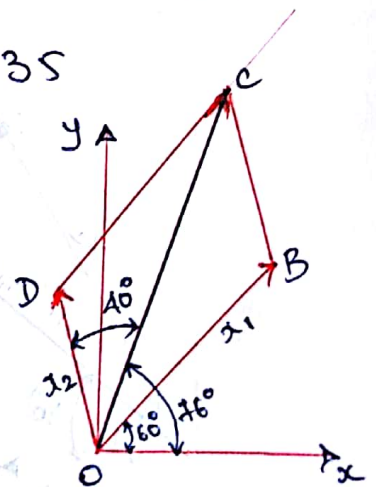
Squaring and adding, we get

$$A = \sqrt{(2.305)^2 + (9.135)^2} = \underline{9.42}$$

$$\tan \alpha = \frac{9.135}{2.305} = 3.963$$

$$\alpha = \tan^{-1} 3.963 = 75.838^\circ$$

$$\therefore x = 9.42 \sin(\omega t + 75.838^\circ)$$



③ Show that the resultant motion of three harmonic motions given below is zero

$$x_1 = a \sin \omega t$$

$$x_2 = a \sin (\omega t + 2\pi/3)$$

$$x_3 = a \sin (\omega t + 4\pi/3)$$

Solution :

$$x = x_1 + x_2 + x_3$$

On solving & expanding.

$$x = a \sin \omega t - a \sin \omega t + 0.866 a \cos \omega t - 0.866 a \cos \omega t$$

$$\boxed{x = 0} \rightarrow \text{Hence resultant motion is zero.}$$

④ A force  $P_0 \sin \omega t$  acts on a displacement  $x_0 \sin (\omega t - \pi/3)$ .  
If  $P_0 = 100 \text{ N}$ ,  $x_0 = 0.02 \text{ m}$ ,  $\omega = 2\pi \text{ rad/sec}$ . Find the work done during i) the first cycle ii) the first second iii) the first quarter second

Solution: Work done = force  $\times$  displacement

$$\therefore W = \int_0^{t_1} P \cdot \frac{dx}{dt} \cdot dt$$

$$\Rightarrow \int_0^{t_1} [P_0 \sin \omega t \cdot \frac{d}{dt} x_0 \sin (\omega t - \pi/3)] dt$$

$$\Rightarrow P_0 x_0 \omega \int_0^{t_1} \sin \omega t \cos (\omega t - \pi/3) dt$$

$$\Rightarrow P_0 x_0 \omega \int_0^{t_1}$$